

Bayesian Decision Making for AI Systems

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1 Introduction

Recent breakthroughs in Artificial Intelligence (AI) and Machine Learning (ML) have revolutionized the way we interact with computers [CTJ⁺21, Ano26], led to significant advances in scientific discovery [JDH⁺23, WFD⁺23, LLL⁺24], and sparked major concerns regarding security, accessibility, and environmental consequences [Hol20, Goo25, BMP⁺25, BCP⁺26]. Against this backdrop, an emerging research literature suggests we should focus not only on developing powerful monolithic foundation models, but also on building *AI systems* based on composable tools providing problem-specific solutions [FZS22, DK24, PAB⁺26]. Examples in software engineering, biological sciences, and AI safety illustrate this position.

1. **Software design.** As a hallmark of current AI capabilities, software design requires the orchestration of multiple basic instructions (e.g., `echo`, `write`, `open`, `grep`, to name only a few) for interacting with the underlying operating system [SDYD⁺23, YZY⁺23, Ano26]. Nonetheless, existing consumer-grade models often struggle to coherently compose these instructions, leading practitioners to consistently rely on closed-source and cloud-based solutions even for simple tasks—such as implementing a function given a set of unit tests.
2. **Drug discovery.** In a similar vein, the *de novo* design of therapeutic drugs has been a major project for the ML community for years [Böh92, GHL20, WFD⁺23]. From a decisional viewpoint, this amounts to defining the pairwise connections between structural motifs maximizing a set of prescribed biological objectives—while abiding by a set of chemical constraints [CC95, CGW12].
3. **AI safety.** Owing to its substantial impact on recent economic progress [Ace24], AI has attracted growing attention from both regulators and policymakers. As such, researchers have increasingly focused on designing policy-compliant models. In practice, the search for policy violations in AI systems and, specifically, in LLMs—known as *red teaming* [PHS⁺22, LKC⁺25, HLD⁺25]—boils down to identifying textual inputs that produce non-compliant outputs, such as NSFW content.

Despite their dissimilarities, these problems share the same fundamental structure: interactive AI models undertaking a *decision problem* over a *combinatorial* space. In (1), a sequence of operations (e.g., read, write, run) should be composed in order to write a test-passing function. In (2), the process is split: one model searches over the space of chemically viable molecular structures, while another assesses the biological properties of each candidate. In (3), an attacker model proposes queries resulting in potentially non-compliant behavior for the target LLM, according to an evaluator network. Let it be code (1), molecules (2), or natural language (3), *compositional discrete objects* are at the core of AI systems.

Beyond their shared combinatorial structure, these problems are also characterized by the presence of uncertain metrics—e.g., flaky tests or inaccurate proxy networks—and inherent multimodality, in which multiple distinct solutions satisfy the same objectives. Crucially, Bayesian theory [BS94, McA98, R⁺07, Bis07] provides a comprehensive and well-established framework for decision making that naturally accommodates these characteristics [MLD⁺23, BCM⁺25, PAB⁺26]. In its generalized form [BHW16, KR24], the Bayesian paradigm assigns both a likelihood, which is based on external information (e.g., unit tests), and a prior probability to each compositional object. These quantities are combined via *Bayes’ rule* [Bay63] to provide a posterior distribution over candidate structures. From this, an intelligent agent can perform inference, execute planning, and

quantify its uncertainty over future outcomes [BS94, BCM⁺25]. There is, however, a fundamental obstacle: the Bayesian approach is NP-hard in general [DC93, DL97].

This has motivated the design of algorithms for carrying out *approximate* Bayesian modeling, many of which date back to the 80’s [GG84, TK86, TW87]. Remarkable landmarks of such development include the Laplace approximation [Mac92], Hamiltonian Monte Carlo [N⁺11], Gibbs sampling [GG84], Langevin dynamics [WT11], and—more recently—diffusion samplers [SDWMG15, HJA20, VGD23]. However, most of these studies have primarily focused on approximating continuous distributions, which dominate both physical and conventional statistical applications. The approximation of discrete distributions over combinatorial spaces, aside from a few ad hoc approaches [RH03, BHK⁺14], remained comparatively less explored until the early 2020s with the introduction of amortized samplers [BJK⁺21, HJE⁺23], which reframed Bayesian inference as a clean-cut learning problem addressable via well-developed deep learning algorithms [LBH15, BB24].

Apart from certain successes in drug discovery [BJK⁺21, JDH⁺23] and natural language processing [HJE⁺23, VJS⁺24], the implementation of these approximated algorithms still challenges both researchers and practitioners, as they are often compute-intensive [LLZB24] and hyperparameter-sensitive [DSCS⁺24]. When compared against their continuous counterparts, existing theoretical frameworks for discrete approximations are also severely underdeveloped, with only a few recent works [KLhO25, SAdSdS⁺25, SSR⁺25] attempting to outline their limitations and capabilities. In this scenario, and given the direction our field is moving towards, the central question guiding the proposed research agenda is

How to efficiently incorporate Bayesian decision making into AI systems?

Research program. Our research plan is grounded on three fundamental axes—theoretical understanding, methodological development, and applications—about which we elaborate next.

1. **Theory.** Drawing on Valiant’s [Val84] and McAllester’s [McA99] PAC and PAC-Bayesian theories and on recent theoretical developments by [DR17, DR18], a number of researchers have provided provable learning guarantees for LLM-scale models [LKA⁺24, LFK⁺24], which partly explain their exceptional capacity. However, corresponding results for approximate Bayesian learners remain scarce, with only a few positive [SSR⁺25] and negative [SAdSdS⁺25] results covering specific empirical configurations. The fundamental question of which Bayesian posteriors can and cannot be efficiently approximated by existing algorithms is largely open.
2. **Methodology.** As mentioned, current approximate Bayesian methods for combinatorial spaces are compute-intensive. In fact, amortized samplers [BJK⁺21, MLD⁺23] and Markov chain Monte Carlo (MCMC) [MRR⁺53, N⁺11] are accurate but suffer from slow convergence, while traditional factorized variational inference algorithms [JGJS99, Bea17] provide a fast but arbitrarily inaccurate approximation. As underlined by the success of adaptive test-time computation in LLMs [WWS⁺23, YYZ⁺23, BBK⁺24], we expect the ideal algorithm to live in between both extremes, allowing for user-specified refinements of an initial approximation at a measurable computational cost with quantifiable statistical improvements.
3. **Applications and Software.** Our theoretical and methodological investigations will also be oriented by specific applications allowing for both fact-checking and benchmarking of the stated claims and designed algorithms. Traditional problems from classical statistics, such as phylogenetic inference [RH03, BHK⁺14] and variable selection [GM93], stand out as potential references with well-established solutions. On the other hand, LLM fine-tuning [HJE⁺23], drug discovery [JDH⁺23], and red-teaming [LKC⁺25] represent tasks with largely open-ended objectives. As we advance, we will also release our developed software under an open-source license to incentivize further research and foster reproducibility.

We summarize these directions into the following conjecture.

Our research program is built upon the conviction that the observed limitations of approximate Bayesian models over combinatorial spaces are not fundamental (1), but rather a consequence of relying on methods originally designed for continuous domains, and that amortized samplers (2) can serve as a principled drop-in replacement for existing ad hoc solutions in both traditional statistical problems and emerging AI applications (3).

We now review prior research we have done (Section 2) and outline future directions we will pursue (Section 3) in lieu of the above objectives.

2 The prior

Our past works have shown the effectiveness of distributed learning [DSCS⁺24, SdSM24, SSR⁺25], introduced provable generalization guarantees [SSR⁺25, SAdSdS⁺25], and characterized the empirical challenges [DdSdS⁺26, DdSC⁺26, SdSdSM24] imposed by training amortized samplers.

Streaming and distributed learning. The posterior distribution in a generalized Bayesian model can often be factorized into easily evaluable components. When observations are independently distributed, for instance, the posterior can be represented as the product of sub-posteriors corresponding to disjoint data subsets [DSCS⁺24]. Similarly, given a partitioned parameter space, the posterior can be represented as the weighted combination of its restrictions to each partition [SSR⁺25]. In both cases, we demonstrated learning can be drastically accelerated by first training an amortized sampler for each component in parallel, and then aggregating the trained samplers in a centralized server [DSCS⁺24, SSR⁺25]. Additionally, we showed a similar technique allows for the training of amortized samplers in dynamic domains—where the target posterior changes over time—without the need for expensive retraining. Importantly, this also enables model reuse and reduces the carbon footprint associated to multiple training runs [SdSM24].

Generalization and expressivity. How close are amortized samplers to the target posterior? This was the primary question we addressed in [SSR⁺25, SAdSdS⁺25]. We first proved that, depending on the chosen architecture and learning algorithm, there are distributions that amortized samplers are fundamentally incapable of approximating [SSR⁺25, SAdSdS⁺25], pioneering the study of their *expressivity*. Correspondingly, we showed conventional assessment protocols in the literature often misrepresented the distributional accuracy of learned samplers, criticizing the purported effectiveness of existing approaches backed solely by brittle evaluation metrics. To address this issue, we introduced a computationally amenable metric and demonstrated—both empirically and theoretically—that it is an accurate surrogate for the otherwise intractable distance of the amortized sampler’s learned distribution to the target posterior. Building on this metric, we derived the first meaningful statistical generalization bounds for amortized samplers—providing quantifiable evidence of their generalization capabilities [SSR⁺25].

Challenges for efficient training. Most recently, we aimed at pinning down the reasons for which training amortized samplers can be computationally challenging. Currently, our findings have underlined three major causes: (i) inadequate state space exploration, (ii) state aliasing, and (iii) redundant computation. As an amortized sampler learns to approximate a posterior distribution by exploring its support, the representativeness and diversity of the encountered regions is crucial for effective learning. To be successful, the learner should swiftly navigate across the target’s high-probability regions. Drawing on this, we developed boosting [DdSdS⁺26] and constrained optimization algorithms [DdSC⁺26] that principledly bias the sampler’s towards high-probability unvisited regions during training—dramatically improving learning efficiency. Additionally, our ongoing works have shown that state aliasing, resulting from a sampler’s failure in distinguishing distinct states due to limited expressivity, is a major bottleneck for obtaining an accurate distributional approximation, which can be alleviated via state space augmentation. Similarly, we have also found that amortized samplers spend significant compute on uninformative states, which are visited countless times throughout training, significantly slowing down convergence. The following

section discusses our current and future efforts to address these and related problems.

3 The posterior

Our research objectives will be developed through the following multi-year projects, whose goal is establish a clear theoretical understanding, promote significant methodological advancements, and pinpoint relevant applications for amortized samplers.

When can amortized samplers be efficiently learned? The astounding success of LLMs can be partly attributed to their simple learning objective—maximum likelihood—and generative process—autoregressive. In fact, under mild assumptions, [Mal24] showed that autoregressive linear models are universal learners when intermediate supervision is available [WWS⁺23]. Amortized samplers, on the other hand, lack even a basic notion of *learnability* that could be used for answering basic questions regarding the difficult of approximating a given distribution. As extensively studied in related topics, such as graph neural networks [KW16, CSJ⁺24] and Gaussian processes [Wil92], which have well-understood limitations and capabilities, our objectives are also to develop a hierarchical description of hardness for amortized samplers, design clear evaluation metrics, and standardize the empirical assessment to foster progress in the field.

Compute-efficient training via shaping. Besides the exploration-, aliasing-, and redundancy-related effects discussed earlier, a fundamental constraint of current amortized samplers is the lack of meaningful intermediate supervision for effective credit assignment. While the differential geometric structure of continuous distributions allows for easier state space navigation by exploiting gradient information, the search for high-probability regions in discrete spaces is essentially dependent upon the black-box dynamics of the chosen learning algorithm. Reward shaping, an concept borrowed from the reinforcement learning literature [NHR99], provides a principled solution to this problem—the central question being how to effectively shape a posterior distribution, which can be interpreted as a reward function. Taken together, reward shaping, state space augmentation, and targeted computation are key techniques to improve the computational tractability and statistical efficiency of amortized samplers.

Large-scale inference through amortization. A distinctive property of amortized samplers is their ability to simultaneously approximate a family of posterior distributions by amortizing over its defining features—for example, over conditioning datasets or the likelihood function’s scale [KKY⁺24]. Recent research, for instance, has investigated training foundation models as a general-purpose solver for Bayesian optimization problems [CLH⁺25, LVB⁺25]. However, the viability of such approach for amortized samplers over combinatorial spaces remains open. As acknowledged by recent research on tabular foundation models [HMEH23, HMP⁺25], a notorious challenge in learning large-scale pre-trained models consists in adequately choosing the training distribution over problems of interest. Curriculum learning [BLCW09] and active inference [Set12], which provide principled and well-established mechanisms for data acquisition in generative modelling, are promising directions for addressing this issue.

4 Teaching Statement

As an educator, I am committed to teaching courses on both foundational and frontier topics in Artificial Intelligence (AI) and Machine Learning (ML), elaborate a modern curriculum that prepares students for their future careers and enhances their critical thinking, and supervise young researchers according to the highest international standards of reproducible and rigorous research. As the saying goes, a research’s major contribution is the young talents they train, advise, and learn from.

Teaching experience. I served as teaching assistant for both undergraduate and graduate (both MSc- and PhD-level) courses on Machine Learning, Linear Algebra, Statistical Inference, Data Analysis, and Time Series. I assisted in designing and grading homeworks and assignments, and held office hours to help students prepare for exams. Additionally, I worked as occasional lecturer

for graduate-level Machine Learning classes on generative modelling and reinforcement learning. I also supervised a group of three undergraduate students in an industry project, helping them to scope a solution for an open-ended problem the company I worked on was facing.

Teaching philosophy. AI and ML lie at the intersection of statistics and computer science, requiring both strong understanding of mathematical foundations and solid hands-on experience with algorithmic design and implementation. These are not contemplative subjects, and can only be learned through repeated practice. As a consequence, providing a list of progressively difficult exercises for students to periodically solve is crucial for their long-term learning. These exercises should cover both pen-and-paper problems, ensuring understanding of core concepts, and lab projects, challenging students to think through the entire ML pipeline.

In addition, graduate students—as researchers in training—need both to acquire a deep comprehension of technical concepts and to understand the tricks of their trade. They should constantly practice writing with clarity, precision, and succinctness. Crucially, they also must know the main venues of their fields, and acknowledge that research papers are not beacons of truth, but rather theses to be challenged.

These directives also provide the guidelines for how to evaluate students. For undergraduate courses, periodic assignments covering recently studied topics, alongside optional exercise sheets, ensure their constant practice and avoid dependence on a single final exam, which would become a single-point-of-failure. Graduate students need fewer written assignments; their focus should be on reading, implementing, and understanding the—often unspoken—limitations of the recent literature. Their findings during this process should be both documented in text and presented to their peers in class, improving their presentation and writing skills with iterated feedbacks.

Teaching interests. I am best suited to teach classes on ML and, in particular, generative modelling and reinforcement learning. However, as someone educated in formal applied mathematics, I am also well-equipped to teach courses on statistics, stochastic processes, measure theory, functional analysis, and related topics.

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